Seminar 07 cheat sheet

Pre-colloquium Practice

Optimization Methods
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Definitions you must know

- **Convex set** C: $tx + (1-t)y \in C$ for every $x, y \in C$ and $t \in [0,1]$.
- **Convex function** on convex C: $f(tx + (1-t)y) \le tf(x) + (1-t)f(y)$.
- **Strongly convex** with $\mu > 0$: $f(y) \ge f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|y-x\|^2$.
- **L-smooth**: $\|\nabla f(y) - \nabla f(x)\| \le L\|y-x\|$ for every x, y.
- **Gradient**: $\nabla f(x) = (\partial_1 f, \dots, \partial_n f)^\top$. **Hessian**: $(\nabla^2 f)_{ij} = \partial_i \partial_j f$, symmetric.
- **Stationary point**: $\nabla f(x^*) = 0$. **Minimum, maximum, saddle** by signs of eigenvalues of $\nabla^2 f(x^*)$.

Problem 3.1: a quadratic form

- $f(x,y) = ax^2 + bxy + cy^2 + dx + ey$: $\nabla f = (2ax + by + d,\ bx + 2cy + e)$.
- $H = \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}$. Solve $Hx^* = -(d,e)^\top$.
- Eigenvalues: $\lambda^2 - \mathrm{tr}(H)\lambda + \det(H) = 0$.
- Both > 0: minimum; both < 0: maximum; different signs: saddle.
- $\mu = \lambda_{\min}(H)$, $L = \lambda_{\max}(H)$ when $H \succ 0$.

Problem 3.2: gradient descent

- $f = \frac{1}{2}x^\top Q x$, $Q \succ 0$: $x_{k+1} = (I - \alpha Q)x_k$.
- Safe: $0 < \alpha < 2/L$. Border: $\alpha = 2/L$ stalls; $\alpha > 2/L$ diverges.
- In the eigenbasis: $e_{k,i} = (1 - \alpha\lambda_i)^k e_{0,i}$.
- $\alpha = 1/L$: steepest axis in one step; slow axis by $1 - \mu/L$.
- Best fixed step $\alpha^* = 2/(\mu + L)$: rate $(\kappa - 1)/(\kappa + 1)$ per step.

Two short proofs

- **Convex combination.** f, g convex, $\alpha, \beta \ge 0$: $h = \alpha f + \beta g$ is convex by linearity in f, g.
- **Strict minimum.** $\nabla f(x^*) = 0$, $\nabla^2 f(x^*) \succ 0$: Taylor gives $f(x^* + h) = f(x^*) + \frac{1}{2}h^\top \nabla^2 f(x^*) h + o(\|h\|^2) > f(x^*)$ for small $h \ne 0$.

What to watch for

- H has $2a$ and $2c$ on the diagonal, not a and c.
- Zero eigenvalue: the second-order test fails, higher-order terms needed.
- $\alpha = 1/L$ is not the best step; $\alpha = 2/(\mu + L)$ is.

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